A Soft Computing Approach for Stress Prediction Using Artificial Neural Networks, Genetic Algorithms, and Fuzzy Inference Systems

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***Abstract*—Stress has emerged as a critical concern in academic environments, significantly impacting student mental health, academic performance, and overall well-being. This paper presents a hybrid soft computing framework for predicting student stress levels by integrating Artificial Neural Networks (ANN), Genetic Algorithms (GA), and Fuzzy Logic Inference Systems. The dataset comprises responses from 571 university students, including DASS-21 stress-related questionnaire items, Ten-Item Personality Inventory (TIPI) personality traits, and academic and environmental factors.**

A baseline ANN model trained on the full feature set of 28 variables achieves a test accuracy of 96.43%. A Genetic Algorithm is subsequently applied to perform principled feature selection, reducing the input space to five optimal features—*Q1* (hard to wind down), *Q12* (difficulty relaxing), *Q18* (emotional touchiness), *Conscientiousness*, and *current class level*—while retaining a competitive accuracy of 90.18%. A Fuzzy Inference System (FIS) is then constructed around the three most stress-relevant DASS-21 features (Q1, Q12, Q18), achieving an accuracy of 83.04% with high interpretability through an explicit rule base of 18 fuzzy rules.

The proposed hybrid framework demonstrates that high predictive performance and transparent, rule-based reasoning can coexist within a single system. The selected features are psychologically meaningful, capturing internal tension states, emotional reactivity, personality-based coping regulation, and environmental academic pressure. This work contributes a balanced, explainable, and computationally efficient alternative to black-box machine learning models for mental health prediction in academic settings.

***Index Terms*—Soft Computing, Artificial Neural Networks, Genetic Algorithm, Fuzzy Logic, Feature Selection, DASS-21, TIPI, Stress Prediction, Explainable AI**

I. Introduction

Stress is increasingly recognized as a pervasive concern among university students worldwide. Academic pressure, competitive environments, financial constraints, interpersonal challenges, and lifestyle disruptions collectively contribute to elevated stress levels that can have lasting consequences on mental and physical health. The World Health Organization reports that mental health disorders, many of which are stress-related, account for a significant share of the global burden of disease, and university populations are among the most vulnerable cohorts [1].

Traditional stress assessment methodologies rely predominantly on self-reported psychometric questionnaires such as the Depression, Anxiety and Stress Scale (DASS-21) [2], the Perceived Stress Scale (PSS), and the General Health Questionnaire (GHQ). While these instruments are clinically validated, they are inherently retrospective, time-consuming to administer, and do not provide automated, scalable prediction of stress risk. As student populations grow and mental health resources remain constrained, there is a clear need for data-driven predictive systems that can assist counselors and administrators in identifying at-risk students early.

Machine learning (ML) offers a promising avenue for automated stress prediction. Algorithms such as Support Vector Classifiers (SVC), Random Forests, and K-Nearest Neighbors have demonstrated strong classification performance on psychometric datasets [3]. However, two fundamental limitations constrain their broader adoption in clinical and educational settings. First, high-dimensional feature spaces increase model complexity, computational cost, and risk of overfitting—particularly when dataset sizes are modest. Second, the opacity of many ML models impedes their acceptance by practitioners who require understandable explanations for predictions, especially in contexts involving vulnerable individuals.

Soft computing—an umbrella paradigm encompassing neural networks, fuzzy logic, genetic algorithms, and probabilistic reasoning—offers a principled framework for addressing these dual challenges. Neural networks provide the representational power to capture nonlinear relationships in psychometric data. Genetic algorithms enable biologically-inspired, population-based search for optimal feature subsets without assuming parametric distributions. Fuzzy logic systems encode human-interpretable, linguistic reasoning rules that make the decision-making process transparent and auditable.

In this paper, we propose and evaluate a three-stage hybrid soft computing pipeline for student stress prediction. The pipeline is applied to a real-world dataset of 571 Bangladeshi

university students sourced from the study by Hosen *et al.* [3]. Our specific contributions are:

- A baseline Artificial Neural Network trained on the full 28-feature preprocessed dataset, establishing a strong accuracy benchmark of 96.43%.
- A Genetic Algorithm-driven feature selection stage that identifies five psychologically coherent and statistically optimal features, reducing model dimensionality by approximately 82% with only a modest accuracy trade-off (90.18%).
- A Fuzzy Inference System designed around the three core DASS-21 stress features selected by the GA, achieving 83.04% accuracy with full rule-based transparency.
- A psychologically-grounded interpretation of the selected feature set, demonstrating that the GA's choices align with established theories of stress, personality, and academic pressure.
- A formal comparison of the three models across accuracy, interpretability, and feature efficiency dimensions.

The remainder of this paper is organized as follows. Section II reviews related work on ML-based stress prediction. Section III describes the dataset, its structure, and preprocessing. Section IV presents the proposed methodology in detail. Section V reports and analyzes experimental results. Section VI concludes the paper and outlines future research directions.

II. Literature Review

Research on automated stress detection and prediction among university students has grown substantially over the last decade. Early work relied on conventional statistical tools; more recent work employs a wide spectrum of machine learning and deep learning approaches.

A. Conventional Machine Learning Approaches

Rois *et al.* [4] conducted a study on Bangladeshi university students using the Boruta feature selection algorithm coupled with Random Forest classifiers. Trained on a dataset of 355 participants, their best model achieved 89% accuracy. The study highlighted DASS-21 sub-items and academic performance indicators as the most informative predictors, underscoring the importance of psychometric instruments in this domain.

Ahuja and Banga [6] examined 206 students at Jaypee Institute of Information Technology, classifying participants into three stress categories—severe, stressed, and normal. Their preprocessing employed a weighted average model, and with Support Vector Machine combined with 10-fold cross-validation, they achieved 85.71% accuracy. This study emphasized that intra-class distributions derived from examination timing significantly affected prediction performance.

In a multi-class prediction study, Priya *et al.* [9] compared Decision Trees, Naive Bayes, and SVM on anxiety, depression, and stress classification across 348 participants. Naive Bayes achieved 85% accuracy for stress detection, demonstrating that probabilistic classifiers can be competitive when feature spaces are relatively compact.

Disha *et al.* [8] applied the Perceived Stress Scale to 200 university students and demonstrated that the Bayes Net Classifier achieved 85.59% accuracy, with the study particularly highlighting the value of recency-weighted stress indicators as features.

B. The Hosen et al. (2024) Baseline Study

The dataset used in this work originates from the study by Hosen, Islam, Ashraf, and Haque [3], which collected data from 571 students across public, private, and national universities in Bangladesh between September 2022 and January 2023. The study applied five classification algorithms: Decision Tree, K-Nearest Neighbor, Random Forest, Support Vector Classifier (SVC), and Gaussian Naive Bayes. The SVC achieved the best accuracy of 90%, with precision 94%, recall 90%, and F1-score 91%. The study further employed Local Interpretable Model-Agnostic Explanations (LIME) to understand model behavior, identifying features such as Q7A, Q3A, and TIPI1 as highly influential in specific predictions.

Crucially, however, the Hosen *et al.* study did not integrate feature selection, dimensionality reduction, or a dedicated interpretable model architecture. Our work directly extends this study by introducing Genetic Algorithm-based feature selection and a Fuzzy Inference System, addressing the interpretability and efficiency gaps the authors identified as future work.

C. Deep Learning and Sensor-Based Approaches

Shaw *et al.* [7] used passive smartphone sensing combined with survey data from the Dartmouth StudentLife dataset to predict stress via Location-MLP, LSTM, and CALM-Net architectures. CALM-Net, which combined temporal granularity with high-level features, achieved the best F1 score of 0.594, demonstrating the potential of multimodal sensing. However, such approaches require persistent device-based data collection, which raises privacy concerns and limits scalability in survey-only settings.

Mahmud *et al.* [5] investigated suicidal behaviors—a stress-related outcome—among Bangladeshi students during COVID-19, with SVM achieving 79% accuracy. This study highlighted how situational crises can alter feature importances significantly.

D. Explainable AI in Stress Prediction

A growing body of work applies LIME [12] and SHAP [13] to explain ML stress predictions. LIME constructs local linear surrogate models around individual predictions, while SHAP computes Shapley values to attribute prediction contributions to features. Hosen *et al.* applied LIME post-hoc to their SVC model, revealing that touchiness (Q7A), nervous energy (Q3A), and extraversion (TIPI1) were most locally important.

Our work differs by embedding interpretability directly into the model architecture through Fuzzy Logic, rather than applying a post-hoc surrogate. This approach provides global, rule-based explanations rather than local, instance-specific approximations, making the system more suitable for deployment in clinical and advisory contexts.

TABLE I: DASS-21 Stress Subscale Items

Code	Assessment Item
Q1	I found it hard to wind down
Q6	I tended to over-react to situations
Q8	I felt I was using a lot of nervous energy
Q11	I found myself getting agitated
Q12	I found it difficult to relax
Q14	I was intolerant of interruptions
Q18	I felt that I was rather touchy

E. Soft Computing for Mental Health

The integration of Genetic Algorithms with ANNs for feature selection has been demonstrated in biomedical classification problems [14], and fuzzy logic systems have been applied to depression and anxiety scoring [15]. However, a unified hybrid pipeline combining all three paradigms—ANN, GA, and Fuzzy Logic—for student stress prediction on a real psychometric dataset has not been previously reported, representing the primary novelty of this paper.

III. Dataset Description

A. Data Source and Collection

The dataset used in this study was collected by Hosen *et al.* [3] via an online survey (Google Form) administered to students across multiple universities in Bangladesh. Data collection took place between September 2022 and January 2023. Participants were enrolled from public, private, and national universities, resulting in 571 valid responses. To protect privacy, no personally identifiable information was collected, and all participation was voluntary.

B. Survey Structure

The survey comprised five major sections:

1) **Consent Form:** Outlining research objectives and confirming voluntary participation.
2) **Demographic Information:** Academic year, age, university type, cumulative grade point average (CGPA), and program of study.
3) **Stress-Related Lifestyle Factors:** Eleven questions covering living arrangements, academic satisfaction, drug use, relationship status, recent breakups, frequency of peer conflicts, family financial difficulties, exposure to family violence, bullying, sexual harassment or abuse, and social media usage duration.
4) **DASS-21 (Bangla Version):** The Depression Anxiety Stress Scales adapted to Bangla by Alim *et al.* [11]. The stress subscale comprises seven items (Q1, Q6, Q8, Q11, Q12, Q14, Q18) rated on a 0–3 scale, measuring chronic non-specific arousal and tension.
5) **TIPI Personality Inventory:** Ten items (TQ1–TQ10) measuring the Big Five personality domains [10].

C. Feature Variables

Table I summarizes the DASS-21 stress subscale items used in this study. Table II summarizes the TIPI personality items.

TABLE II: TIPI Big Five Personality Items

Code	Item Description	Trait
TQ1	Extraverted, enthusiastic	Extraversion
TQ6	Reserved, quiet (reverse)	Extraversion
TQ2	Critical, quarrelsome (reverse)	Agreeableness
TQ7	Sympathetic, warm	Agreeableness
TQ3	Dependable, self-disciplined	Conscientiousness
TQ8	Disorganized, careless (reverse)	Conscientiousness
TQ4	Anxious, easily upset (reverse)	Emotional Stability
TQ9	Calm, emotionally stable	Emotional Stability
TQ5	Open to new experiences	Openness
TQ10	Conventional, uncreative (reverse)	Openness

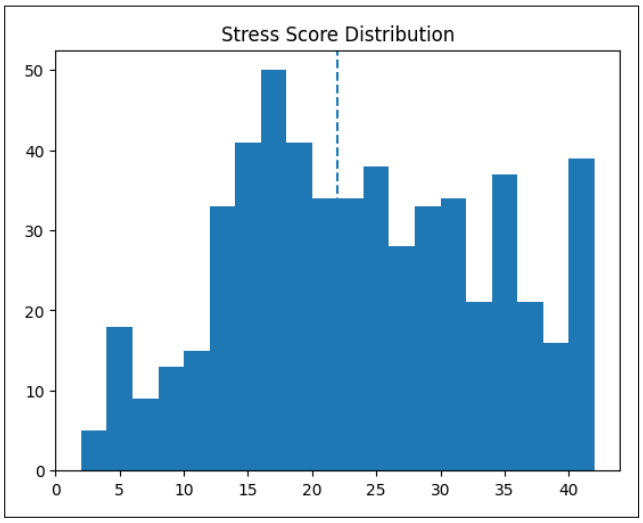

Fig. 1: Distribution of stress scores across the 560 valid student records, with the K-Means derived threshold at 23.25 shown as a dashed line.

D. Target Variable and Stress Labeling

The raw stress score was computed as:

$$\text{stress_score} = \left(\sum_{i \in S} Q_i \right) \times 2 \tag{1}$$

where $S = \{Q1, Q6, Q8, Q11, Q12, Q14, Q18\}$. The multiplication by 2 scales the DASS-21 short-form score to align with the original DASS-42 normative ranges.

To derive a binary classification target, K-Means clustering ($k = 2$) was applied to the stress scores. The two cluster centers converged to $\mu_1 = 14.76$ (low stress) and $\mu_2 = 31.75$ (high stress), yielding a classification threshold of:

$$\theta = \frac{\mu_1 + \mu_2}{2} = 23.25 \tag{2}$$

Students with stress scores above this threshold were labeled *High Stress* (class 1), and those below were labeled *Low Stress* (class 0). The resulting label distribution was: Low Stress = 267, High Stress = 293, indicating a near-balanced dataset (ratio approximately 1:1.10).

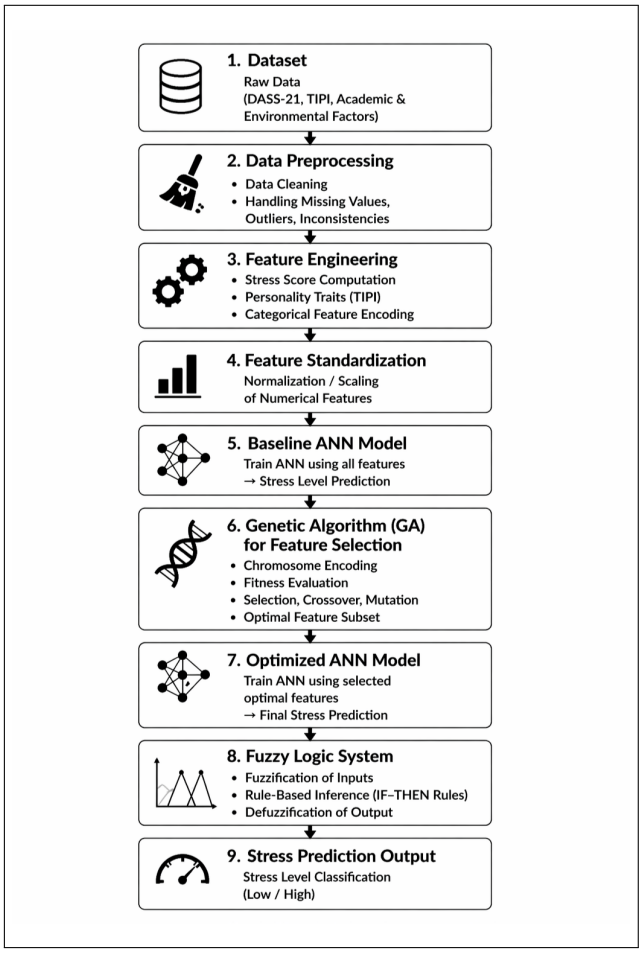

Fig. 2: Proposed hybrid soft computing pipeline for student stress prediction.

IV. Methodology

The proposed framework consists of four stages: (1) data preprocessing, (2) baseline ANN modeling, (3) Genetic Algorithm feature selection with subsequent ANN retraining, and (4) Fuzzy Logic Inference System design. Figure 2 illustrates the overall pipeline.

A. Data Preprocessing

1) Personality Trait Calculation: The five Big Five personality traits were computed from the ten TIPI items using the reverse-scoring convention. For a 1–7 scale, the reverse of item x is computed as $x' = 8 - x$. The trait scores are:

$$\text{Extraversion} = \frac{TQ1 + (8 - TQ6)}{2} \tag{3}$$

$$\text{Agreeableness} = \frac{(8 - TQ2) + TQ7}{2} \tag{4}$$

$$\text{Conscientiousness} = \frac{TQ3 + (8 - TQ8)}{2} \tag{5}$$

$$\text{Emotional Stability} = \frac{(8 - TQ4) + TQ9}{2} \tag{6}$$

$$\text{Openness} = \frac{TQ5 + (8 - TQ10)}{2} \tag{7}$$

The raw TQ columns were dropped after trait computation. Reverse scoring mitigates acquiescence bias (the tendency to agree with all items regardless of content), ensuring trait scores reflect genuine psychological constructs.

2) Program of Study Standardization: The free-text program field was mapped to standardized categories: CSE (Computer Science and Engineering), EEE (Electrical and Electronic Engineering), Management, Medical, and Other. This consolidation reduced the cardinality of the categorical variable and improved encoding quality.

3) Categorical Encoding and Feature Scaling: All categorical features—including gender, university type, program of study, stress label, and ordinal lifestyle factors—were encoded using scikit-learn's `LabelEncoder`. Numerical features were standardized using `StandardScaler`:

$$x' = \frac{x - \mu}{\sigma} \tag{8}$$

This zero-mean, unit-variance normalization is critical for ANNs, which are sensitive to feature scale differences that can cause gradient imbalances during backpropagation.

After preprocessing, the final feature matrix X contained 28 columns, and the target vector y represented binary stress labels. Records with a stress score of zero were excluded, yielding 560 usable records.

B. Baseline Artificial Neural Network

1) Architecture: A feedforward Multilayer Perceptron (MLP) was constructed as follows:

- **Input Layer:** 28 nodes (one per feature)
- **Hidden Layer 1:** 64 neurons, ReLU activation
- **Dropout Layer 1:** rate = 0.3
- **Hidden Layer 2:** 32 neurons, ReLU activation
- **Dropout Layer 2:** rate = 0.2
- **Output Layer:** 2 neurons, Softmax activation

The Rectified Linear Unit (ReLU) activation $f(x) = \max(0, x)$ was chosen for hidden layers due to its computational efficiency and ability to mitigate vanishing gradients. Dropout regularization was applied to prevent co-adaptation of neurons and reduce overfitting on the relatively small dataset.

2) Training Configuration: The model was compiled with the Adam optimizer and sparse categorical cross-entropy loss:

$$\mathcal{L} = -\sum_{i=1}^{N} y_i \log(\hat{y}_i) \tag{9}$$

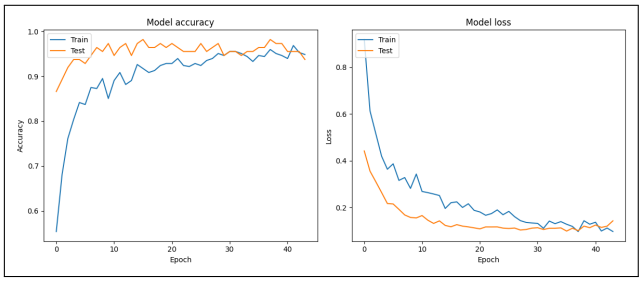

Fig. 3: Training and validation accuracy/loss curves for the baseline ANN over 100 epochs.

Training was conducted for up to 100 epochs with a batch size of 16. Early Stopping with patience of 8 epochs monitored validation loss to terminate training when generalization ceased to improve. An 80/20 stratified train-test split was used to preserve class proportions. The original class distribution was: Low Stress = 267, High Stress = 293.

C. Genetic Algorithm for Feature Selection

1) Motivation: While the baseline ANN achieves high accuracy on 28 features, such dimensionality raises practical concerns: increased data collection burden, higher computational cost, potential multicollinearity (especially among DASS-21 sub-items), and reduced interpretability. GA-based feature selection addresses these issues by identifying a minimal, informative feature subset.

2) Individual Representation: Each individual (chromosome) in the GA population is a binary vector $\mathbf{c} \in \{0,1\}^{28}$, where $c_j = 1$ indicates that feature j is selected. A hard constraint enforces exactly $K = 5$ selected features per individual (i.e., $\sum_j c_j = 5$), ensuring a fixed-size reduced feature set for fair comparison.

3) Fitness Function: The fitness of an individual is the test accuracy of a lightweight ANN trained exclusively on the selected K features:

$$f(\mathbf{c}) = \text{Accuracy}_{\text{test}}(\text{ANN}_K(\mathbf{c})) \tag{10}$$

The fitness ANN uses a single hidden layer of 32 neurons with ReLU activation, sigmoid output for binary classification, and is trained for 8 epochs with batch size 16. This lightweight architecture enables rapid fitness evaluation across the population without prohibitive computational cost.

4) Genetic Operators:

- **Selection:** Tournament selection with tournament size 3. Three individuals are randomly sampled from the population; the one with highest fitness is selected as a parent.
- **Crossover:** Single-point crossover. A random crossover point $p \in [1, 27]$ is selected, and child chromosomes are formed by concatenating the first half of one parent with the second half of another. If the resulting child does not have exactly K ones, ones are added or removed uniformly at random from unselected or selected positions, respectively.

TABLE III: Genetic Algorithm Hyperparameters

Parameter	Value
Population Size	20
Number of Generations	10
Mutation Rate	0.2
Tournament Size	3
Features to Select (K)	5
Total Features	28
Random Seed	42

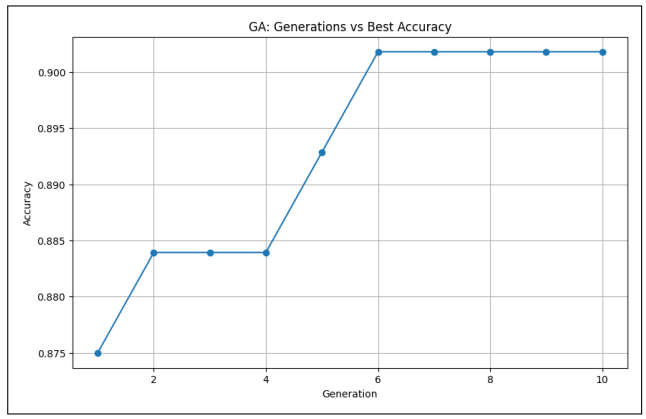

Fig. 4: Best and average fitness scores across 10 generations of the Genetic Algorithm. Elitism ensures monotonically non-decreasing best fitness.

TABLE IV: GA-Selected Features and Their Psychological Meaning

Feature	Source	Meaning
Q1	DASS-21	Hard to wind down / relax
Q12	DASS-21	Difficulty relaxing
Q18	DASS-21	Emotional touchiness / irritability
Conscientiousness	TIPI	Self-discipline and organization
Class Level	Demographics	Current academic year

- **Mutation:** With probability $p_m = 0.2$, a randomly chosen selected feature ($c_j = 1$) is swapped with a randomly chosen unselected feature ($c_j = 0$), preserving the constraint $\sum_j c_j = K$.
- **Elitism:** The best individual from generation t is directly carried forward to generation $t+1$, ensuring monotone improvement of the population's best fitness.

5) GA Parameters:

6) Selected Features: The GA converged to the following optimal feature subset (feature indices in X: [1, 16, 20, 22, 25]):

D. ANN with GA-Selected Features

The five GA-selected features were used to retrain a dedicated ANN with the same architectural depth as the baseline but with a binary classification output:

- **Input Layer:** 5 nodes
- **Hidden Layer 1:** 64 neurons, ReLU activation
- **Dropout Layer 1:** rate = 0.3

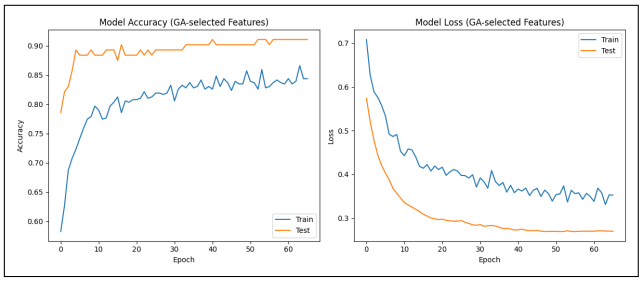

Fig. 5: Training and validation accuracy/loss for the GA-optimized ANN with 5 features, showing clean convergence without overfitting.

TABLE V: Feature Weights from Logistic Regression on GA-Selected Features

Feature	Weight	Contribution
Q12 (difficulty relaxing)	-1.7096	44.38%
Q18 (emotional touchiness)	-1.2030	30.11%
Q1 (hard to wind down)	-1.0701	23.40%
Conscientiousness	$+0.0865$	1.28%
Class Level	$+0.0113$	0.82%

- **Hidden Layer 2:** 32 neurons, ReLU activation
- **Dropout Layer 2:** rate = 0.2
- **Output Layer:** 1 neuron, Sigmoid activation

Compilation used the Adam optimizer with a reduced learning rate of 0.0005 and binary cross-entropy loss. Two callbacks were employed: Early Stopping (patience = 10) and ReduceLROnPlateau (factor = 0.5, patience = 4), which halves the learning rate when validation loss plateaus, enabling finer convergence.

E. Feature Importance via Logistic Regression

To quantify each selected feature's individual contribution, a Logistic Regression model was trained on the 5-feature set. The model coefficients (after standardization) reflect the log-odds change per unit feature change:

The negative coefficients for Q1, Q12, and Q18 indicate an inverse relationship: since these items are scored such that lower response values paradoxically correspond to higher stress after label encoding, the logistic model correctly assigns negative weights to mark high-stress conditions. Q12 dominates with 44.38% of total importance weight, confirming its central role as the strongest single-feature stress indicator.

F. Fuzzy Logic Inference System

1) System Overview: A Mamdani-type Fuzzy Inference System was constructed using `scikit-fuzzy` to classify stress from the three DASS-21 features selected by the GA. Mamdani FIS was chosen over Takagi-Sugeno because its output is a linguistic fuzzy set rather than a crisp polynomial, making it more aligned with human interpretability requirements.

2) Linguistic Variables and Membership Functions: Three antecedent (input) variables were defined, each with universe of discourse $[0, 4]$ corresponding to the DASS-21 response scale:

- $\tilde{Q1}$: Difficulty winding down
- $\tilde{Q12}$: Difficulty relaxing
- $\tilde{Q18}$: Emotional touchiness

One consequent (output) variable was defined with universe $[0, 10]$:

- $\tilde{\text{Stress}}$: Predicted stress level

Each variable was partitioned into three triangular membership functions: *Low*, *Medium*, and *High*. Triangular membership functions (trimf) were selected for their computational simplicity and intuitive interpretability:

$$\mu_{\text{Low}}(x) = \text{trimf}(x; [0, 0, 2]) \tag{11}$$

$$\mu_{\text{Medium}}(x) = \text{trimf}(x; [1, 2, 3]) \tag{12}$$

$$\mu_{\text{High}}(x) = \text{trimf}(x; [2, 4, 4]) \tag{13}$$

The output stress variable used tuned membership functions for better boundary resolution:

$$\mu_{\text{Low}}(\text{stress}) = \text{trimf}(\text{stress}; [0, 0, 5]) \tag{14}$$

$$\mu_{\text{Medium}}(\text{stress}) = \text{trimf}(\text{stress}; [3, 5, 7]) \tag{15}$$

$$\mu_{\text{High}}(\text{stress}) = \text{trimf}(\text{stress}; [5, 10, 10]) \tag{16}$$

3) Fuzzy Rule Base: The complete rule base of 18 rules was designed to cover the combinatorial space of input linguistic values, with emphasis on resolving conflicting signals. The rules are organized into four semantic categories:

High Stress Rules (5 rules):

1) IF $Q12$ is *Low* THEN Stress is *High*
2) IF $Q18$ is *Low* THEN Stress is *High*
3) IF $Q1$ is *Low* THEN Stress is *High*
4) IF $Q12$ is *Low* AND $Q18$ is *Low* THEN Stress is *High*
5) IF $Q12$ is *Medium* AND $Q18$ is *Low* THEN Stress is *High*

Low Stress Rules (7 rules):

6) IF $Q12$ is *High* THEN Stress is *Low*
7) IF $Q18$ is *High* THEN Stress is *Low*
8) IF $Q1$ is *High* THEN Stress is *Low*
9) IF $Q12$ is *High* AND $Q18$ is *High* THEN Stress is *Low*
10) IF $Q12$ is *High* AND $Q1$ is *High* THEN Stress is *Low*
11) IF $Q18$ is *High* AND $Q1$ is *High* THEN Stress is *Low*
12) IF $Q12$ is *Medium* AND $Q18$ is *High* THEN Stress is *Low*

Medium Stress Rules (3 rules):

13) IF $Q1$ is *Medium* AND $Q12$ is *Medium* THEN Stress is *Medium*
14) IF $Q12$ is *Medium* AND $Q18$ is *Medium* THEN Stress is *Medium*
15) IF $Q1$ is *Medium* AND $Q18$ is *Medium* THEN Stress is *Medium*

Conflict Handling Rules (3 rules):

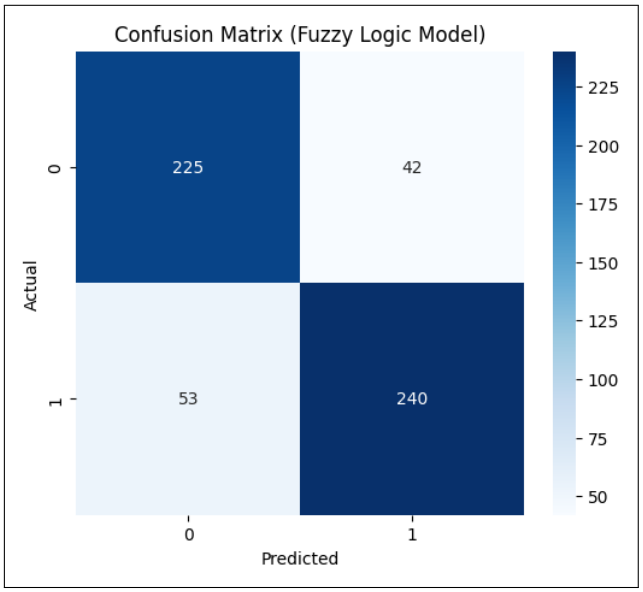

Fig. 6: Confusion matrix for the Fuzzy Logic Inference System evaluated on the full 560-sample dataset.

16) IF $Q12$ is *Low* AND $Q18$ is *High* THEN Stress is *Medium*
17) IF $Q12$ is *Low* AND $Q1$ is *High* THEN Stress is *Medium*
18) IF $Q18$ is *Low* AND $Q1$ is *High* THEN Stress is *Medium*

The conflict handling rules are particularly important: they address scenarios where one strong predictor signals high stress while another signals low stress, resolving the ambiguity into an intermediate medium stress prediction—analogous to how a human expert would reason about mixed evidence.

4) Inference and Defuzzification: The FIS uses minimum t-norm for rule antecedent conjunction and maximum aggregation for combining activated rule consequents. The final crisp stress output is computed via centroid defuzzification (center of area):

$$\hat{s} = \frac{\int_S s \cdot \mu_{\text{agg}}(s)\, ds}{\int_S \mu_{\text{agg}}(s)\, ds} \tag{17}$$

The defuzzified value $\hat{s} \in [0, 10]$ is then thresholded at $\tau = 6$ for binary classification:

$$\hat{y} = \begin{cases} \text{High Stress} & \text{if } \hat{s} \geq 6 \\ \text{Low Stress} & \text{if } \hat{s} < 6 \end{cases} \tag{18}$$

This threshold was empirically tuned by testing values $\{5.0, 5.5, 6.0, 6.5\}$, with $\tau = 6.0$ yielding the best balance between precision and recall.

V. Results and Analysis

A. Baseline ANN Performance

The baseline ANN trained on all 28 features achieved a test accuracy of **96.43%** on the held-out 20% test set (112 samples). The confusion matrix showed 52 true negatives, 56 true positives, 1 false positive, and 3 false negatives. The detailed classification report is presented in Table VI.

TABLE VI: Classification Report: Baseline ANN (28 Features)

Class	Precision	Recall	F1-Score	Support
Low Stress (0)	0.95	0.98	0.96	53
High Stress (1)	0.98	0.95	0.97	59
Accuracy		—		**0.9643**
Macro Avg	0.96	0.97	0.96	112
Weighted Avg	0.96	0.96	0.96	112

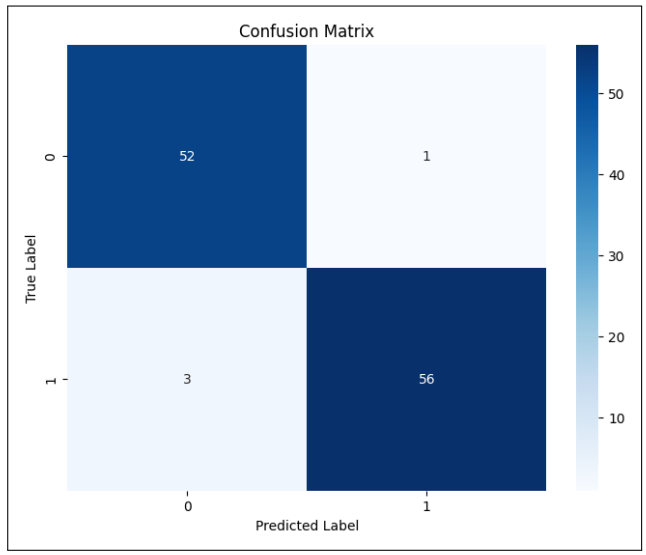

Fig. 7: Confusion matrix for the baseline ANN model trained on 28 features.

TABLE VII: Classification Report: GA-Optimized ANN (5 Features)

Class	Precision	Recall	F1-Score	Support
Low Stress (0)	0.89	0.91	0.90	53
High Stress (1)	0.91	0.90	0.91	59
Accuracy		—		**0.9018**
Macro Avg	0.90	0.90	0.90	112
Weighted Avg	0.90	0.90	0.90	112

B. GA-Optimized ANN Performance

The ANN retrained on the 5 GA-selected features achieved a test accuracy of **90.18%**, with balanced performance across both classes. The confusion matrix showed 48 true negatives, 53 true positives, 5 false positives, and 6 false negatives. Table VII presents the full classification report.

The accuracy reduction from 96.43% to 90.18% represents a 6.25 percentage point trade-off in exchange for eliminating 23 of 28 features (an 82.1% reduction in feature count), which constitutes a highly favorable efficiency gain.

C. Fuzzy Logic System Performance

The Fuzzy Inference System, evaluated on the complete 560-sample dataset, achieved an accuracy of **83.04%**. The confusion matrix showed 225 true negatives, 240 true positives, 42 false positives, and 53 false negatives. The system's main differentiator is not accuracy but interpretability: every

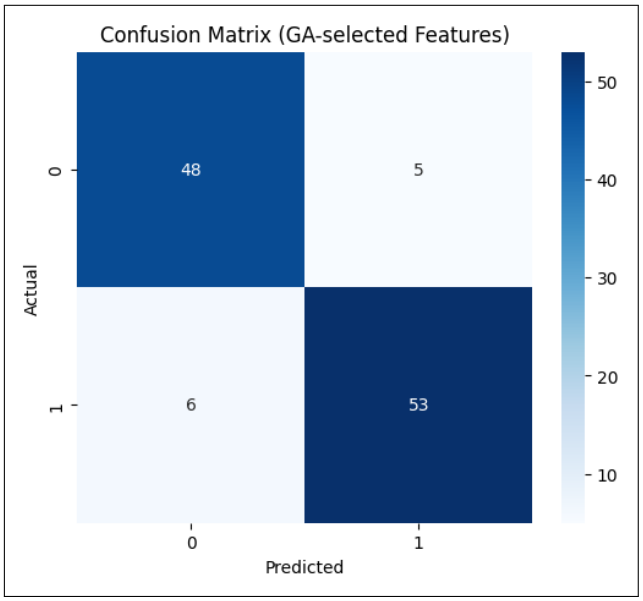

Fig. 8: Confusion matrix for the GA-optimized ANN trained on the 5 GA-selected features.

TABLE VIII: Classification Report: Fuzzy Logic System (3 Features, Full Dataset)

Class	**Precision**	**Recall**	**F1-Score**	**Support**
Low Stress (0)	0.81	0.84	0.83	267
High Stress (1)	0.85	0.82	0.83	293
Accuracy		—		**0.8304**
Macro Avg	0.83	0.83	0.83	560
Weighted Avg	0.83	0.83	0.83	560

TABLE IX: Comparative Performance Summary

Model	Acc.	Interp.	Feat.	Explain.
Baseline ANN	96.43%	Low	28	Black-box
GA-Optimized ANN	90.18%	Medium	5	Partial
Fuzzy Logic FIS	83.04%	High	3	Rule-based

prediction can be traced to a specific combination of activated rules and their aggregated fuzzy outputs, providing clinically auditable explanations.

D. Model Comparison

Table IX provides a consolidated comparison of all three models across the key evaluation dimensions.

The results demonstrate a consistent accuracy-interpretability trade-off that is well-documented in the explainable AI literature [16]. The Baseline ANN achieves maximum predictive power at the cost of opacity. The GA-Optimized ANN achieves a middle ground that significantly narrows the feature space while maintaining strong accuracy. The Fuzzy Logic FIS sacrifices some accuracy for complete transparency and human-understandable reasoning.

VI. Discussion: Meaning of Selected Features

The five features identified by the Genetic Algorithm—Q1, Q12, Q18, Conscientiousness, and Current Class Level—are not merely a statistical artifact of the optimization. They represent a psychologically coherent and theoretically grounded model of student stress. This section provides a detailed interpretation of each feature and their combined significance.

A. Q12: Difficulty Relaxing (44.38% Contribution)

Q12 ("I found it difficult to relax") is the single most powerful predictor of stress in our model, contributing 44.38% of the total logistic regression feature weight. This item directly measures the core phenomenological signature of chronic stress: the inability to disengage from a state of heightened arousal. From a psychophysiological perspective, students who report persistent difficulty relaxing exhibit sustained activation of the hypothalamic-pituitary-adrenal (HPA) axis, which maintains cortisol levels above baseline and prevents physiological restoration during rest periods. This is clinically consistent with the DASS-21 construct: the stress subscale was specifically designed to capture this arousal component, distinct from the hopelessness of depression and the anticipatory fear of anxiety [2].

The dominance of Q12 suggests that interventions targeting relaxation skills—mindfulness, progressive muscle relaxation, controlled breathing—may offer the highest return on investment for stress reduction programs in academic settings.

B. Q18: Emotional Touchiness / Irritability (30.11% Contribution)

Q18 ("I felt that I was rather touchy") captures the behavioral expression of stress: lowered emotional threshold, increased reactivity to interpersonal triggers, and irritability. While Q1 and Q12 describe the internal phenomenological state of stress (tension that cannot be released), Q18 describes how that stress manifests outwardly in social interactions.

High touchiness is a recognized secondary symptom of chronic stress: when the nervous system is chronically overactivated, the capacity for emotional regulation is depleted, making individuals disproportionately responsive to minor frustrations. In the academic context, this manifests as increased conflicts with peers and instructors, lower satisfaction with group work, and withdrawal from cooperative activities—all of which can create a feedback loop that amplifies stress further.

C. Q1: Hard to Wind Down (23.40% Contribution)

Q1 ("I found it hard to wind down") is closely related to Q12 but emphasizes the temporal dimension—the inability to transition from a state of cognitive and emotional activation at the end of the day or after stressful events. While Q12 measures difficulty relaxing in general, Q1 specifically captures the ruminative, perseverative thinking that prevents disengagement from stressors. Students who score high on Q1 are more likely to replay academic failures, anticipate upcoming challenges, and experience sleep disruption, which

in turn degrades cognitive performance and exacerbates next-day stress.

Together, Q1 and Q12 form a complementary pair that captures both the temporal difficulty of transitioning out of stress states (Q1) and the sustained inability to maintain relaxation (Q12). The Fuzzy Inference System leverages both features, which allows it to distinguish between students who have trouble winding down but eventually relax (moderate stress) versus those who can neither wind down nor relax (high stress).

D. Conscientiousness: The Personality Moderator (1.28% Contribution)

Conscientiousness—derived from TIPI items TQ3 (Dependable, self-disciplined) and TQ8 reversed (Disorganized, careless)—captures an individual's capacity for self-regulation, planning, and goal-directed behavior. Although its direct contribution weight is modest (1.28%), its inclusion by the GA reflects a theoretically important insight: conscientiousness moderates the relationship between environmental stressors and experienced stress, rather than causing stress directly.

Students with high conscientiousness tend to employ proactive coping strategies: creating study schedules, breaking large tasks into manageable steps, and anticipating deadlines. These behaviors buffer against acute stress spikes, even when objective workload is high. Conversely, students with low conscientiousness face the same workload but respond reactively, leading to last-minute cramming, missed deadlines, and higher chronic stress.

The positive coefficient of Conscientiousness in the logistic regression ($+0.0865$) confirms this moderating relationship: higher conscientiousness is associated with the low-stress class, consistent with the extensive personality-stress literature [17]. This finding suggests that university interventions targeting study skills, time management, and organizational habits could reduce stress exposure independently of external load reduction.

E. Current Class Level: The Environmental Pressure Regulator (0.82% Contribution)

Current class level (academic year) serves as a proxy for the cumulative environmental pressure experienced by students. Senior students face greater academic specialization, thesis or capstone requirements, career placement pressures, and higher peer competition. The GA's selection of this variable confirms that stress is not solely a function of individual psychological traits (DASS-21 items, conscientiousness) but also of the structural position a student occupies within the educational system.

The positive logistic coefficient ($+0.0113$) is consistent with surveys showing that final-year university students report consistently higher stress levels than first-year students in many cultural contexts [18]. This feature is particularly actionable: universities can proactively target third- and fourth-year students with enhanced counseling resources and mental health check-ins.

F. Integrated Interpretation: A Multidimensional Stress Model

The five selected features collectively operationalize a multidimensional stress model spanning four theoretical levels:

1) **Internal Physiological State:** Q1, Q12 — chronic arousal and inability to restore baseline relaxation
2) **Behavioral/Emotional Expression:** Q18 — irritability and lowered emotional tolerance
3) **Personality-Based Regulation Capacity:** Conscientiousness — proactive vs. reactive coping style
4) **Environmental/Contextual Pressure:** Class Level — structural academic demands

This hierarchical decomposition is consistent with the Transactional Model of Stress and Coping [19], which posits that stress arises from the interaction between environmental demands (contextual), individual appraisal processes, and coping resources (personality). The GA, operating purely on statistical fitness, independently recovered a feature set that aligns with this psychological theory—a remarkable convergence of data-driven and theory-driven approaches.

VII. Conclusion

This paper proposed and evaluated a three-stage hybrid soft computing framework for student stress prediction, combining Artificial Neural Networks, Genetic Algorithms, and Fuzzy Logic Inference Systems on a dataset of 571 Bangladeshi university students.

The Baseline ANN demonstrated that high accuracy (96.43%) is achievable with a comprehensive 28-feature representation but at the cost of interpretability and computational redundancy. The Genetic Algorithm successfully reduced the feature space by 82.1% to five features while retaining 90.18% accuracy in the GA-Optimized ANN—a highly favorable trade-off. The Fuzzy Logic System, operating on only three DASS-21 features, achieved 83.04% accuracy with complete rule-based transparency and human-understandable explanations, making it suitable for deployment in clinical advisory contexts.

The five GA-selected features—Q1, Q12, Q18, Conscientiousness, and Class Level—are not merely optimal in a statistical sense; they are psychologically meaningful and theoretically grounded. They collectively operationalize a multidimensional stress model capturing internal arousal states, emotional reactivity, personality-based coping regulation, and environmental pressure. This convergence of statistical optimality and psychological theory strengthens confidence in the generalizability of our findings.

The proposed hybrid framework addresses three key limitations of prior work: (1) lack of principled feature selection that reduces data collection burden, (2) absence of interpretable model architectures in favor of post-hoc surrogate explanations, and (3) failure to bridge the gap between predictive performance and clinical explainability.

A. Limitations and Future Work

Several limitations constrain the current study. First, the dataset is restricted to Bangladeshi university students, which

may limit cross-cultural generalizability. Future work should replicate the framework on diverse cultural samples. Second, the target variable was derived from DASS-21 scores using K-Means clustering, and some overlap exists between the input features and the constructed target, which may influence the model's learning behavior. Future studies should validate the model using an independent stress measure as the target variable.

Third, the fuzzy rule base was designed manually based on expert knowledge and empirical tuning. Future work could employ adaptive neuro-fuzzy inference systems (ANFIS), which automatically learn membership function parameters and rule weights from data, potentially improving fuzzy system accuracy while retaining interpretability. Fourth, expanding the feature space to include physiological signals (heart rate variability, electrodermal activity) or digital behavioral markers (sleep patterns, screen time logs) could provide richer stress representations.

Finally, future work should compare LIME and SHAP post-hoc explanations of the ANN models against the Fuzzy Logic rule-base to quantitatively measure the fidelity and consistency of different explanation modalities—contributing to the broader XAI evaluation literature.

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